

Data Analyst – Quiz Question Bank

60 Multiple-Choice Questions with Correct Answers

Question 1. Which SQL clause is used to filter rows before grouping?

- A. WHERE
- B. HAVING
- C. GROUP BY
- D. ORDER BY

Correct Answer: A

Question 2. Which SQL function returns the number of rows?

- A. SUM()
- B. COUNT()
- C. TOTAL()
- D. ROWS()

Correct Answer: B

Question 3. Which SQL clause is used to group rows with the same values?

- A. ORDER BY
- B. GROUP BY
- C. WHERE
- D. DISTINCT

Correct Answer: B

Question 4. Which JOIN returns all rows from the left table and matching rows from the right table?

- A. INNER JOIN
- B. RIGHT JOIN
- C. LEFT JOIN
- D. CROSS JOIN

Correct Answer: C

Question 5. Which SQL keyword removes duplicate rows from a result?

- A. UNIQUE
- B. DISTINCT
- C. REMOVE
- D. FILTER

Correct Answer: B

Question 6. Which SQL command is used to retrieve data?

- A. SELECT
- B. FETCHDB

- C. GET
- D. OPEN

Correct Answer: A

Question 7. Which aggregate function calculates the average?

- A. MEAN()
- B. AVG()
- C. AVERAGE()
- D. MID()

Correct Answer: B

Question 8. Which clause filters grouped results after aggregation?

- A. WHERE
- B. HAVING
- C. GROUP BY
- D. LIMIT

Correct Answer: B

Question 9. Which SQL operator is used for pattern matching?

- A. MATCH
- B. LIKE
- C. PATTERN
- D. SEARCH

Correct Answer: B

Question 10. Which constraint uniquely identifies each row in a table?

- A. FOREIGN KEY
- B. PRIMARY KEY
- C. CHECK
- D. DEFAULT

Correct Answer: B

Question 11. Which Excel function counts cells containing numbers?

- A. COUNTA
- B. COUNT
- C. COUNTIF
- D. SUM

Correct Answer: B

Question 12. Which Excel feature summarizes and analyzes large datasets quickly?

- A. PivotTable
- B. WordArt
- C. Mail Merge

D. Goal Seek

Correct Answer: A

Question 13. Which Excel function looks up a value vertically in a table?

- A. VLOOKUP
- B. HLOOKUP
- C. FIND
- D. SEARCH

Correct Answer: A

Question 14. Which Excel function can return a value based on a matching position?

- A. INDEX
- B. SUM
- C. ROUND
- D. TEXT

Correct Answer: A

Question 15. Which Excel function counts cells that meet a condition?

- A. COUNTIF
- B. COUNTA
- C. COUNTNUM
- D. SUMIF

Correct Answer: A

Question 16. Which Excel function adds values based on a condition?

- A. SUMIF
- B. ADDIF
- C. TOTALIF
- D. COUNTIF

Correct Answer: A

Question 17. What is the purpose of conditional formatting?

- A. Change formulas
- B. Highlight cells based on rules
- C. Create databases
- D. Delete duplicates

Correct Answer: B

Question 18. Which chart is commonly used to show trends over time?

- A. Pie chart
- B. Line chart
- C. Scatter-free chart
- D. Gauge only

Correct Answer: B

Question 19. Which Excel tool can split text into multiple columns?

- A. Text to Columns
- B. Data Merge
- C. Split Rows
- D. Column Filter

Correct Answer: A

Question 20. What does a PivotTable primarily provide?

- A. Data encryption
- B. Data summarization
- C. Code compilation
- D. Email automation

Correct Answer: B

Question 21. Which Python library is commonly used for tabular data analysis?

- A. Pandas
- B. Flask
- C. Requests
- D. Tkinter

Correct Answer: A

Question 22. Which Pandas structure is two-dimensional and tabular?

- A. Series
- B. DataFrame
- C. ArrayList
- D. Dictionary

Correct Answer: B

Question 23. Which Pandas function reads a CSV file?

- A. read_csv()
- B. open_csv()
- C. csv_read()
- D. load_table()

Correct Answer: A

Question 24. Which Pandas method displays the first rows of a DataFrame?

- A. head()
- B. first()
- C. top()
- D. start()

Correct Answer: A

Question 25. Which Pandas method removes rows containing missing values?

- A. dropna()
- B. removeNull()
- C. deleteNA()
- D. clearna()

Correct Answer: A

Question 26. Which Pandas method replaces missing values?

- A. fillna()
- B. replaceNull()
- C. fixna()
- D. setna()

Correct Answer: A

Question 27. Which Python library is widely used for numerical computation?

- A. NumPy
- B. Django
- C. BeautifulSoup
- D. PyTest

Correct Answer: A

Question 28. What does a NumPy array primarily provide?

- A. Structured numerical data
- B. Email services
- C. Web routing
- D. SQL storage

Correct Answer: A

Question 29. Which Pandas method provides summary statistics?

- A. describe()
- B. summary()
- C. stats()
- D. profile()

Correct Answer: A

Question 30. Which Pandas operation combines data based on matching keys?

- A. merge()
- B. joinText()
- C. combineRows()
- D. attach()

Correct Answer: A

Question 31. What is data cleaning?

- A. Improving data quality by correcting issues
- B. Encrypting files
- C. Creating passwords
- D. Installing databases

Correct Answer: A

Question 32. What is a missing value in a dataset?

- A. A value that is not present or recorded
- B. A duplicate column
- C. A calculated total
- D. A chart label

Correct Answer: A

Question 33. What is an outlier?

- A. An unusually distant observation

- B. A duplicate row
- C. A column name
- D. A data type

Correct Answer: A

Question 34. Which technique can help detect outliers using quartiles?

- A. IQR method
- B. Binary search
- C. Hashing
- D. Tokenization

Correct Answer: A

Question 35. What does data normalization generally aim to do?

- A. Put data into a consistent scale or structure
- B. Delete all data
- C. Create passwords
- D. Compress images

Correct Answer: A

Question 36. What is a duplicate record?

- A. A repeated record representing the same entity or event
- B. A calculated field
- C. A unique key
- D. A chart

Correct Answer: A

Question 37. Why is data validation important?

- A. To ensure data meets defined rules and quality requirements
- B. To increase screen brightness
- C. To install Python
- D. To create slides

Correct Answer: A

Question 38. What is ETL?

- A. Extract, Transform, Load
- B. Evaluate, Test, Link
- C. Export, Transfer, List
- D. Encode, Transform, Login

Correct Answer: A

Question 39. What is the purpose of data transformation?

- A. Convert data into a useful and consistent format
- B. Delete databases

C. Create user accounts

D. Install software

Correct Answer: A

Question 40. Which measure represents the middle value of ordered data?

A. Mean

B. Median

C. Mode

D. Range

Correct Answer: B

Question 41. Which measure is the most frequently occurring value?

- A. Mean
- B. Median
- C. Mode
- D. Variance

Correct Answer: C

Question 42. What does standard deviation measure?

- A. Dispersion of values around the mean
- B. Number of columns
- C. Database size
- D. Missing values

Correct Answer: A

Question 43. What does correlation measure?

- A. The strength and direction of association between variables
- B. The number of rows
- C. The database password
- D. The median only

Correct Answer: A

Question 44. What is a KPI?

- A. Key Performance Indicator
- B. Known Processing Input
- C. Key Programming Interface
- D. Kernel Performance Index

Correct Answer: A

Question 45. What is descriptive analytics?

- A. Analysis that summarizes what happened
- B. Analysis that guarantees the future
- C. Database encryption
- D. Software deployment

Correct Answer: A

Question 46. What is predictive analytics?

- A. Analysis used to estimate future outcomes using data and models
- B. Formatting an Excel sheet
- C. Deleting historical data
- D. Creating passwords

Correct Answer: A

Question 47. What is a dashboard?

- A. A visual interface that presents key data and metrics
- B. A database table only
- C. A programming language
- D. A file compression tool

Correct Answer: A

Question 48. Which Power BI component is commonly used to create interactive reports?

- A. Power BI Desktop
- B. Power BI Compiler
- C. Power BI Writer
- D. Power BI Terminal

Correct Answer: A

Question 49. What is a slicer in Power BI?

- A. An interactive filter for report data
- B. A database server
- C. A Python package
- D. A file format

Correct Answer: A

Question 50. What is DAX primarily used for in Power BI?

- A. Creating calculations and measures
- B. Managing operating systems
- C. Writing HTML pages
- D. Sending emails

Correct Answer: A

Question 51. Which visualization is suitable for comparing categories?

- A. Bar chart
- B. Line chart only
- C. Map only
- D. Gauge only

Correct Answer: A

Question 52. Which visualization is commonly used to show parts of a whole?

- A. Pie chart
- B. Scatter plot
- C. Line chart
- D. Histogram

Correct Answer: A

Question 53. Which visualization is useful for showing the relationship between two numeric variables?

- A. Scatter plot

- B. Pie chart
- C. Card
- D. Treemap only

Correct Answer: A

Question 54. What is a data warehouse?

- A. A centralized system designed for analytical data storage
- B. A web browser
- C. A spreadsheet formula
- D. A programming IDE

Correct Answer: A

Question 55. What is a fact table in a star schema?

- A. A table containing measurable business events or facts
- B. A table containing only passwords
- C. A user interface
- D. A configuration file

Correct Answer: A

Question 56. What is a dimension table?

- A. A table containing descriptive attributes used to analyze facts
- B. A table containing only calculations
- C. A server log only
- D. A Python module

Correct Answer: A

Question 57. What is data granularity?

- A. The level of detail represented in the data
- B. The number of passwords
- C. The file extension
- D. The database engine

Correct Answer: A

Question 58. Why should analysts document data sources?

- A. To improve traceability, understanding, and reproducibility
- B. To increase CPU speed
- C. To remove SQL syntax
- D. To change screen resolution

Correct Answer: A

Question 59. What is an analytical dashboard's main purpose?

- A. Communicate important metrics and insights for decision-making
- B. Store source code only

C. Replace a database

D. Encrypt all files

Correct Answer: A

Question 60. What is a primary benefit of automation in data analysis?

A. It reduces repetitive manual work and improves consistency

B. It guarantees every prediction is correct

C. It removes the need for validation

D. It eliminates all errors

Correct Answer: A